



Applying inherent capabilities of quantum-dot cellular automata to design: D flip-flop case study

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ABSTRACT

Nowadays, quantum cellular automata (QCA) has been considered as the pioneer technology in next generation computer designs. QCA provides the computer computations at nano level using molecular components as computation units. Although the QCA technology provides smaller chip area and eliminates the spatial constraints than earlier CMOS technology, but different characteristics and design limitations of QCA architectures have led to essential attentions in replacement of traditional structures with QCA ones. Inherent information flow control, limited wire length, and consumed area are of such features and restrictions. In this paper, D flip-flop structure has been considered and we have proposed two new D flip-flop structures which employ the inherent capabilities of QCA in timing and data flow control, rather than ordinary replacement of CMOS elements with equivalent QCA ones. The introduced structures involve small number of cells in contrast to earlier proposed ones in presence of the same or even lower input to output delay. The proposed structures are simulated using the QCADesigner and the validity of them has been proved.

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1. Introduction

Quantum cellular automata (QCA) is an emerging nanotechnology that has been gaining attention in the research community due to its extremely small feature sizes, ultra low power consumption, simple concept, and potential for real world applications in the near future. QCA is based on the interaction of bi-stable QCA cells constructed from four quantum-dots as shown in Fig. 1. Each cell is constructed from four quantum-dots arranged in a square pattern. The cell is charged with two electrons, which are free to tunnel between adjacent dots. These electrons tend to occupy antipodal sites as a result of their mutual electrostatic repulsion. Thus, there exist two equivalent energetically minimal arrangements of the two electrons in the QCA cell as shown in Fig. 1. These two arrangements are denoted as cell polarization $P = +1$ and $P = -1$, respectively. By using cell polarization $P = +1$ to represent logic “1” and $P = -1$ to represent logic “0”, binary information can be encoded.

QCA has gained significant popularity in recent years. This is mainly due to rising interest in creating computing devices at the nano-scale. At such scales, QCA has an inherent advantage over conventional integrated circuit technologies (such as CMOS) in that its performance increases as feature sizes are reduced, whereas CMOS exhibits decreasing performance with decreasing feature

sizes. Investigations into modeling and design have demonstrated that QCA has many powerful features some of which are not available in CMOS [1,2].

Recently, a lot of efforts have been performed to develop QCA based logics. In Khurashia et al. and Fijany et al. [4,5], primary structure of a one-bit adder has been proposed and in Wang et al. [6] the authors have modified the adder to an optimum one with small number of cells. The reference [3] is about a multiple-bit adder using carry lookahead. Paper [16] presents the design, layout, and successful simulation of a configurable logic block (CLB) for a field-programmable gate array (FPGA) architecture based on quantum-dot cellular automata. A quantum-dot cellular automata complex gate has been proposed in Townsend and Abraham [12] which composed of simple 3-input majority gates. This 7-input gate can be configured into many useful gate structures such as a 4-input AND gate, a 4-input OR gate, a product of sums representation, a sum of products representation, and other variations. Some simple QCA based processors are also available in literature [11,14]. Some other works include digital ram [8,9], QCA based FPGA [10], serial adder [11], Barrel shifter [3], and And-Or-Inverter (AOI) complex gate [18].

Authors of Ravichandran et al. [13] have developed the first cell-level placement of QCA circuits, where the given circuit is assumed to be partitioned into 4-phase asynchronous QCA timing zones. They have formulated the QCA cell placement in each timing zone as a unidirectional geometric embedding of k-layered bipartite graphs. In that paper, an analytical and a stochastic solution has been presented for minimizing the wire crossings and wire length

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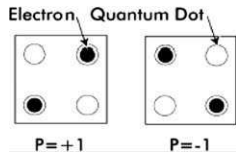


Fig. 1. QCA cells' polarization.

in these placement solutions. Limited wire length is one of the most important design issues in QCA. As demonstrated in Kim et al. [15], noise amplification can be prevented by either limiting length of the wires or synchronizing the flow of signals to let them arrive on time. In addition, some other design notes are discussed in Kim et al. [15].

The design and characterization of sequential circuits in QCA have not been fully addressed in the technical literature. While sequential elements can be implemented using QCA memory cells [20], such an approach would be very slow in performance [19]. Some earlier methods (such as [19]), have considered the implicit latching in QCA, exploiting the four-phase clocking scheme. But this results in complexity of sequential circuit design due to timing constraints. Regarding such limitations and also, using specific features of QCA, we have proposed two novel methods for implementation of low latency edge triggered D flip-flop that demonstrate independent clock input. Independent clock input reduces the attention to timing constraints and simplifies the design of sequential circuits in QCA. This knowledge may also be helpful for FSM design in QCA circuits.

This paper has been organized as follows. In part 2, we will introduce the basic components and logics in QCA. Section 3 is about data flow control and clocking zones in QCA. In Section 4, traditional structures of D flip-flop are investigated and two novel methods are introduced. In Section 5, simulation results are shown and compared to traditional QCA based D flip-flop architectures from consumed area and delay view points. Finally the paper will be concluded in Section 6.

2. Fundamental logics in QCA

In this part, the fundamental logics and gates are investigated in the order of wire, inverter, majority gate, AND and OR gates.

2.1. QCA wire

Fig. 2 illustrates how a binary value propagates down the length of a QCA wire [1]. In this figure, the wire is a horizontal row of QCA cells. The binary signal propagates from left-to-right because of the Coulombic interactions between cells. We assume that charges in cell 1 are trapped in polarization $P = -1$ but those in cells 2–9 are not trapped. Hence, there is no danger that the wire could reverse directions and have a polarization propagate in the direction from which it came. The binary value of 0 (polarization $P = -1$) will propagate down the length of the wire because of the Coulombic interactions between cells. Initially, the electron repulsion caused by Coulombic interaction between cell 1 and cell 2 will cause cell

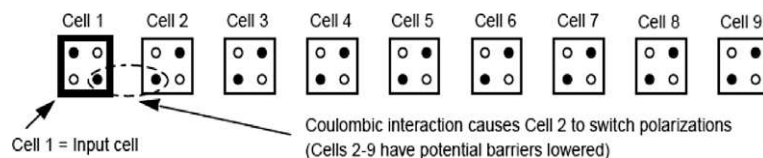


Fig. 2. QCA wire structure [11].

2 to change polarizations. Then, the electron repulsion between cell 2 and cell 3 will cause cell 3 to change polarizations. This process will continue down the length of the QCA wire.

A QCA wire can also be comprised of cells oriented at 45-degrees as opposed to the 90-degree orientation discussed above where the binary value propagates down the length of the wire while it alternates between polarization $P = +1$ and polarization $P = -1$ along the wire cells.

2.2. Inverter gate

There are multiple ways to do it in the QCA paradigm that one is shown in Fig. 3. The signal comes from the left, splits into two parallel wires, and is inverted at the point of convergence. The design is geometrically symmetric, so inversion of a 1 or a 0 occurs with the same reliability. We have mentioned earlier that there is no meaning to talk about coming in of a signal, as there are no currents here. But if we fix the polarization of one of the cells then the symmetry of the cells adjacent to this cell is broken. The only degree of freedom is the one associated with the driver cell.

2.3. Majority, AND, and OR gates

The majority gate is shown in Fig. 4, respectively. The majority gate performs a three-input logic function. Assuming the inputs are A, B and C, the logic function of the majority gate is

$$M(A, B, C) = AB + AC + BC \quad (1)$$

By fixing the polarization of one input to logic “1” or “0”, we can obtain an OR gate and an AND gate, respectively, as illustrated in (2),

$$\begin{aligned} M(A, B, 0) &= AB + (B0) + (A0) = AB \\ M(A, B, 1) &= AB + (B1) + (A1) \\ &= AB + B + A = B + A \end{aligned} \quad (2)$$

More complex logic circuits can then be constructed from OR and AND gates.

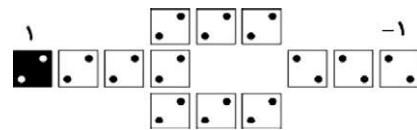


Fig. 3. A QCA inverter gate.

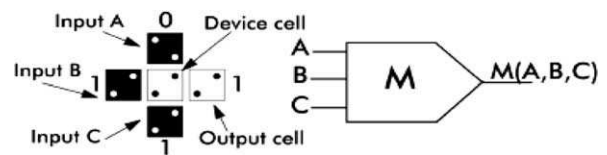


Fig. 4. Majority gate structure [6].

3. Data flow control and clock zones

Two types of cells' state switching are possible in the operation of QCA where so called adiabatic switching is usually preferred. In adiabatic switching, a clock signal is utilized to ensure adiabatic switching of QCA cells. The clock signals for QCA are generated through an electric field that is applied to the cells to either raise, or lower the *tunneling barrier* between dots (which is called inter-dot barrier) within a QCA cell. When the barrier is low, the cells are in a non-polarized state; when the barrier is high, the cells are not allowed to change state. Adiabatic switching is achieved by lowering the barrier, removing the previous input, applying new input to the cell and then raising the barrier [21].

The clock in QCA is multi-phased and includes four clocking zones. During the first clock phase, the switch phase, the inter-dot potential barriers are low. The barriers are then raised during this phase and the QCA cells become polarized according to the state of their driver (i.e. their input cell). It is in this clock phase that the actual computation occurs. By the end of this clock phase, barriers are high enough to suppress any electron tunneling and cell states are fixed. During the second clock phase, the hold phase, barriers are held high so the outputs of the sub-array can be used as inputs to the next stage. In the third clock phase, the release phase, barriers are lowered and cells are allowed to relax to an unpolarized state. Finally, during the fourth clock phase, the relax

phase, cell barriers remain lowered and cells remain in an unpolarized state [11]. The four clock phases and their effect on cells have been illustrated in Fig. 5.

Fig. 6 shows the electrical potential of inter-dot barriers when applying the clock signal. Considering the data flow control, suppose an array of QCA cells which has been divided into sub-arrays where each sub-array is in one of the four clocking zone. It means that for each sub-array, a single potential modulates the inter-dot barriers of all cells simultaneously. The clock signal of neighboring sub-arrays lags or leads in one clocking phase. This clocking scheme allows one sub-array to perform a certain calculation, having its state frozen by the raising of its inter-dot barriers' potential. In the next clock zone, the output of that sub-array act as the input to the successor sub-array to change its state. Along the calculation phase (switch phase) of one sub-array, the successor sub-array is kept in an unpolarized state, so it will not be influenced from the current calculation. Fig. 7 shows five consequent cells in an array where each cell is in a distinct clocking zone. This array of cells has been considered during five time steps and the electrical field of first cell has been shown during these time steps. As can be seen, a binary value of 0 has been propagated from left-to-right of the array which has been indicated by a dark line along the switching phase.

4. D flip-flop structure

Since the similarity of the latch to the RAM cell in terms of functionality is quite obvious, the RAM cell design which is introduced in Walus et al. [8] gives rise to a latch design followed by a flip-flop design. The stripped down version of the RAM cell which acts as a gated D-latch is shown in Fig. 8.

The QCA cell layout diagram which is shown in Fig. 9, involves 50 cells and four clocking zones; hence input–output delay is a single clock cycle long.

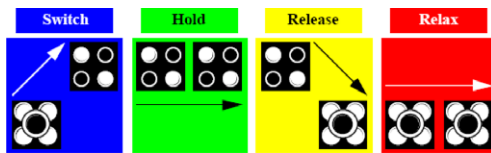


Fig. 5. Four continuous clocking zones.

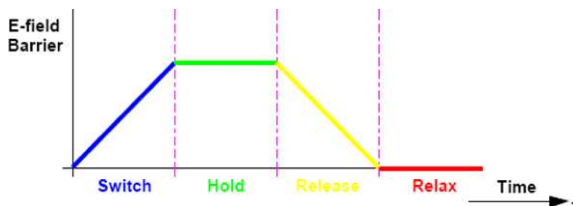


Fig. 6. The electrical potential of inter-dot barriers during four clocking zones.

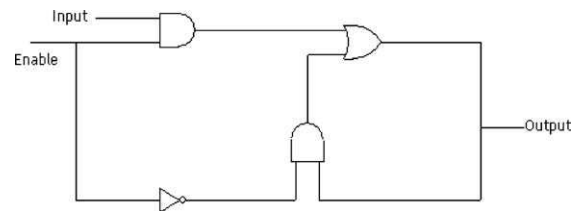


Fig. 8. D-latch circuit diagram.

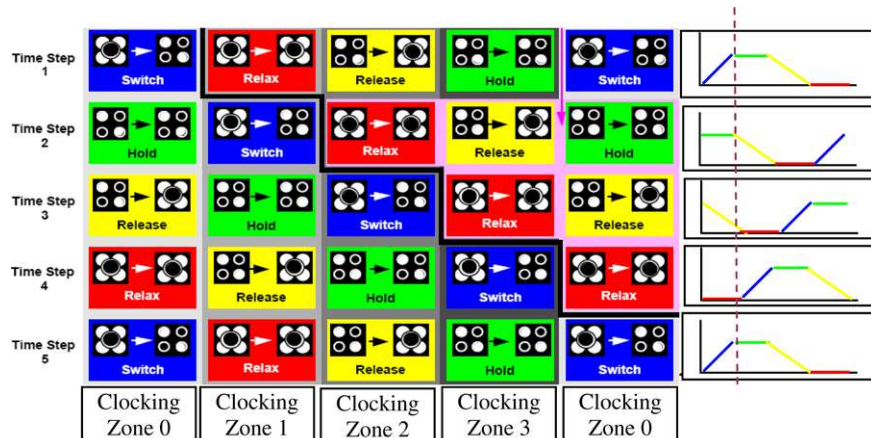


Fig. 7. Propagation of binary data along a QCA wire with five cells.

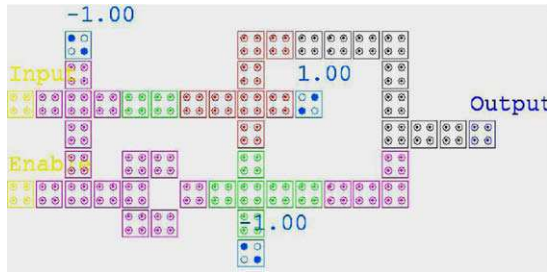


Fig. 9. QCA layout of D-latch.

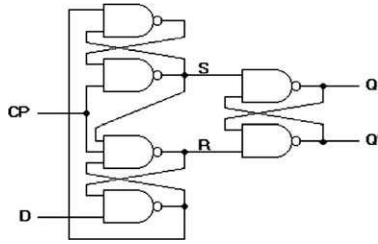


Fig. 10. Traditional CMOS D flip-flop.

4.1. Traditional QCA based D flip-flop

There are two different ways flip-flops arise in traditional circuitry. The first one is illustrated in Fig. 10 and is the circuit layout of a negative edge triggered D flip-flop. We cannot construct an analogue of this design using QCA as we have no analogue of cross coupled NAND gates.

The other method of traditional CMOS implementations is the master-slave combination of two latches which is possible in QCA. The QCA circuit layout for the master-slave arrangement is shown in Fig. 11 [4]. The layout clearly shows two latches cascaded one after the other.

This design uses 133 QCA cell while the wasted area includes about 400 cells. It spans over eight clocking zones from input to output and so the input–output delay of the circuit is two clock cycles.

In Huang et al. [19], authors have proposed an RS flip-flop which provides the output after six clocking zones. This structure covers an area of about 200 cells and uses about 70 cells as shown in Fig. 12. This architecture would require additional logics, cells and subsequent delay to be converted to an edge triggered D flip-flop with independent clock input.

Authors of Huang et al. [19] have indicated that D flip-flops can be implemented using binary wires in QCA where input signal is delivered to the output after at least one complete clock cycle delay (four zones). Although this is acceptable regarding the clocking phases in QCA data flow, but timing constraints and synchronization of components must be considered in sequential circuit design. Timing constraints come from the fact that in QCA, the four-phase clock signal controls not only the flip-flops, but also the combinational gates. Therefore, we must treat an appropriate assignment of clocking zones to QCA circuits such as the algorithm proposed in Huang et al. [19]. In later subsections, we are going to introduce novel structures for edge triggered D flip-flop that demonstrate independent clock input with optimized delay and cell utilization. This independent clock input reduces the effect of timing constraints and simplifies the design of sequential circuits.

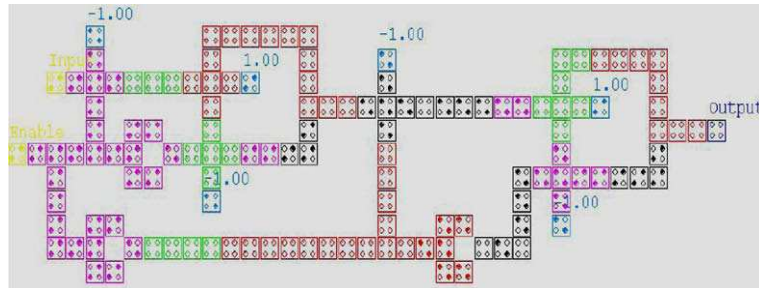


Fig. 11. QCA layout of a master-slave D flip-flop.

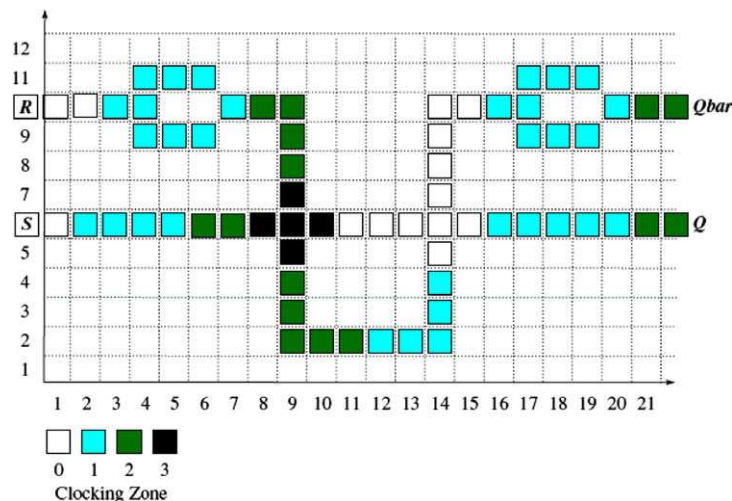


Fig. 12. The proposed RS flip-flop in Huang et al. [19].

4.2. The first proposed D flip-flop

As discussed earlier, the master-slave approach viewed in Section 4.1 consists of about 130 cells and requires eight clocking zones to produce the output. As illustrated in literature, long wire paths and wasted area are major obstacles in QCA implementation. In this part, we propose an edge triggered D flip-flop that utilizes smaller number of cells and space compared to previous ones. It involves a D-latch and an “edge to level converter” as shown in Fig. 13.

As shown in Fig. 14, the proposed converter unit utilizes the inherent capability of clocking zones in QCA. An AND operation is applied between input clock and delayed version of its bit reversed. When the input clock transits from low to high, (+1) will be raised in the output since the bit reverse of CLK lags four clocking zones. The truth table of edge to level converter is available in Table 1. In the latch unit, the input data will be latched when the Enable signal arises. The edge to level converter activates Enable input of D-latch when the CLK input raises from low to high. Hence, the proposed scheme in Fig. 13 behaves as a positive edge triggered D flip-flop.

4.3. The second proposed novel structure

As described for earlier structure in previous section, although the number of cells has been diminished, the output of D flip-flop is still available after eight clocking zones. In this section, we propose another innovative structure which diminishes the above delay, in addition to its smaller number of cells. This structure has been shown in Fig. 15. As indicated in this scheme, the structure includes four majority gates and one inverter. Delay units are no more than additional four clocking zones in the CLK path as shown by numbers in this figure. The validity of proposed scheme has been proved in Table 2.

As can be seen in Table 2, there are four distinct states which are discussed as below:

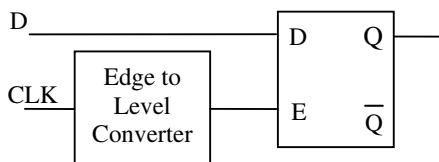


Fig. 13. Proposed architecture for D flip-flop.

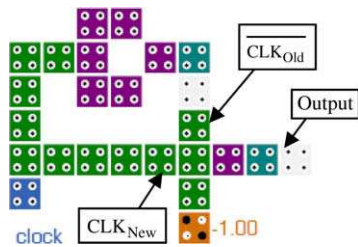


Fig. 14. QCA layout of the edge to level converter.

Table 1

Edge to level conversion truth table.

CLK status	CLK _{New}	$\overline{CLK}_{Old}$	Output
1	1	–1	–1
–1	–1	1	–1
–1 to 1 Transition	1	1	1
1 to –1 Transition	–1	–1	–1

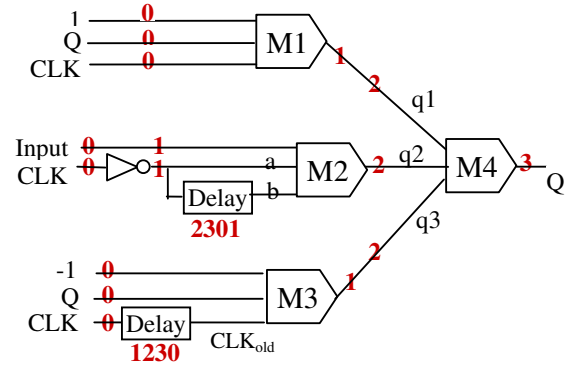


Fig. 15. Second approach's structure; four clocking zones are shown as 0, 1, 2 and 3.

- (a) When the CLK is +1 continuously:
 - The inputs of M1 majority gate in Fig. 15 are as +1, Q , +1 and so its output, $q1$, becomes +1, respectively.
 - In M2 majority gate, the values of a and b are both –1 which results that output $q2$ becomes –1.
 - The inputs of the M3 majority gate are –1, Q and +1 which results that output of this gate, $q3$, becomes equal to Q .
 - As above, providing +1, –1 and Q as inputs of M4 majority gate, its output becomes equal to Q (this means that Q remains as its earlier one).
- (b) When the CLK is –1 continuously:
 - The inputs of M1 majority gate are as +1, Q , –1 and so its output, $q1$, becomes Q , respectively.
 - In M2 majority gate, the values of a and b are both +1 which results that output $q2$ becomes +1.
 - The inputs of the M3 majority gate are –1, Q and –1 which results that output of this gate, $q3$, becomes equal to –1.
 - As above, providing Q , +1 and –1 as inputs of M4 majority gate, its output is equal to Q (Q remains in its earlier value).
- (c) When the CLK transits from +1 to –1:
 - The inputs of M1 majority gate are as +1, Q , –1 and so its output, $q1$, becomes equal to Q .
 - In M2 majority gate, the value of a is +1 and the value of b is –1 and so the inputs of M2 are $Input$, +1 and –1 which results that output $q2$ becomes equal to $Input$.
 - In M3, CLK_{Old} is +1 and so the inputs of M3 majority gate are –1, Q and +1 which results that output of this gate, $q3$, becomes equal to Q .
 - Providing Q , $Input$ and Q as inputs of M4 majority gate, its output becomes equal to Q (Q remains as its earlier one).
- (d) When the CLK transits from –1 to +1 (raising edge):
 - The inputs of M1 majority gate are as +1, Q , +1 and so its output, $q1$, becomes +1, respectively.
 - In M2 majority gate, the value of a is –1 and the value of b is +1 and so the inputs of M2 are $Input$, –1 and +1 which results that output $q2$ becomes equal to $Input$.

Table 2

Second approach's truth table.

CLK status	CLK _{Old}	CLK _{now}	$q1$	$q2$	$q3$	Q
–1	–1	–1	Q	1	–1	Q
1 to –1 transition	1	–1	Q	$Input$	Q	Q
–1 to 1 transition	–1	1	1	$Input$	–1	$Input$
1	1	1	1	–1	Q	Q

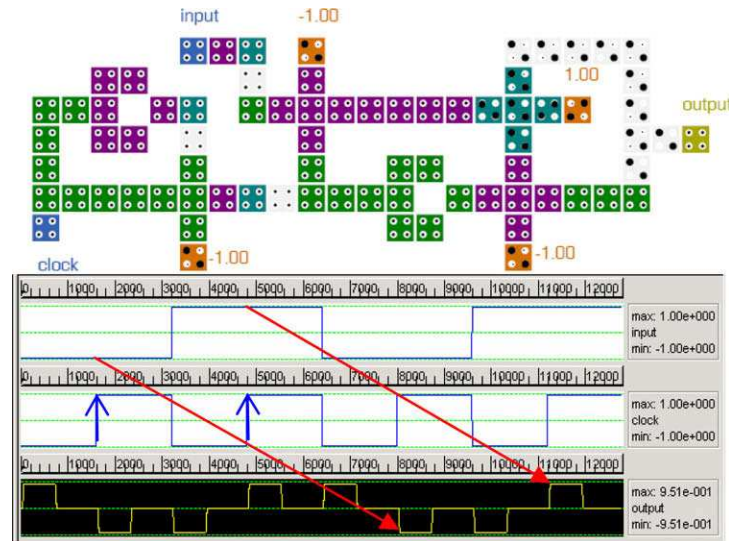


Fig. 16. The simulation and its results for the first proposed method; the output appears after eight clocking zones.

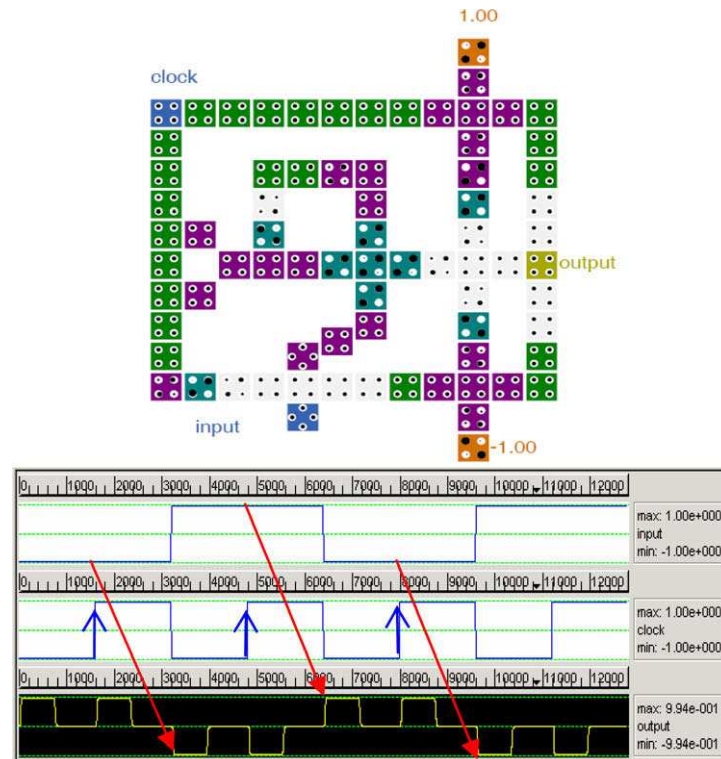


Fig. 17. The simulation and its results for the second proposed method; the output appears after four clocking zones.

- In M3, CLK_{old} is -1 and so the inputs of M3 majority gate are -1 , Q and -1 which results that output of this gate, $q3$, becomes -1 .
- Providing $+1$, $Input$ and -1 as inputs of M4 majority gate, its output becomes equal to $Input$ (Q only change under this condition to the value of $Input$).

5. Simulation results

In this section the proposed methods are simulated and their validity has been proved using QCADesigner. The QCADesigner is

a simulator which simulates the QCA components and reports the state of input–output signals as the result. The results are compared in terms of the number of consumed cells, the wasted area, and the input to output delay.

The implementation of the first proposed model is shown in Fig. 16. As seen, the number of cells in this implementation is around 76 cells and the wasted area covers about 180 cells. Red arrows¹ in the chart show the input to output mapping of binary

¹ For interpretation of color in Figs. 16 and 17, the reader is referred to the web version of this article.

Table 3

Comparison of different implementations.

	Number of cells	Wasted area in cells	Input to output delay
Traditional master-slave D flip-flop	133	407	Eight clocking zones
First proposed method	76	184	Eight clocking zones
Second proposed method	75	168	Four clocking zones

data. Just like the traditional method, the delay from applying CLK input to preparing the valid output is eight clocking zones (or two clock cycles).

The second proposed method has been simulated as shown in Fig. 17. The number of used cells is 75 ones and the wasted area involves about 170 cells. Meanwhile, the major advantage over the earlier proposed method is in its input to output delay which is four clocking zones. The red arrows show the input to output mapping of binary data. The amount of delay is equivalent to one clock cycle which is one clock cycle less than earlier method. Table 3 shows the final comparison of mentioned methods.

As you can see in this table, although the first proposed method reduces the number of used cells comparing to traditional master-slave flip-flop, it yet prepares the output after eight clocking zones. In the second proposed method, the output appears after four clocking zones while the number of cells has been diminished too. These results show that our second proposed model is as fast as a D flip-flop that has been implemented using binary wires in QCA, while in our model the input clock is provided independently of inherent clocking zones. This may be helpful in designing sequential circuits than using binary wires regarding its timing and synchronization constraints.

Nevertheless, reducing the number of cells and required clocking zones would burden some operating constraints such as working frequencies. Using *Bistable Simulation* mode in QCADesigner, these restrictions are not so obvious for the first proposed method. However, the second proposed approach works properly only when the “number of samples” option is above 2000 and below 14,600. Discussing about the reason behind this fact is not in the scope of this paper and will be considered as future works.

6. Conclusion

As discussed, some design features such as inherent data flow control and some design limitations such as limited wire length and consumed area have led to growing attentions in replacement of CMOS structures with QCA ones. In QCA implementation of equivalent CMOS circuits, it is essential to consider the inherent characteristics of QCA technology instead of ordinary replacement of basic gates with the majority gates. The capability of developing precise timing of data flow must be noted as one of those inherent features in QCA. In this paper, we have proposed two innovative methods for a D flip-flop structure with the mentioned features. Both of the methods are more beneficial in terms of used cells and input to output delay, considering simulation results.

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